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Agency: Networking and Information Technology Research and Development (NITRD) National Coordination Office (NCO) on Behalf of the Office of Science and Technology Policy (OSTP)

Re: Request for Information (RFI) on the Development of an Artificial Intelligence (AI) Action Plan

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Introduction

SEMI member companies are grateful to OSTP for its leadership and for maintaining a productive dialogue with industry. SEMI members welcome the opportunity to provide the following comments in response to the RFI on the development of an AI Action Plan, per Executive Order 14179.

Founded in 1970, SEMI is the leading global industry association dedicated to advancing the electronics manufacturing supply chain. With more than 3,000 members worldwide—including more than 650 U.S.-headquartered companies—SEMI represents a broad spectrum of the semiconductor industry, including chip designers, equipment manufacturers, materials suppliers, and subcomponent producers. While SEMI's membership includes large multinational corporations, more than 85 percent of its members are small and medium-sized enterprises (SMEs). Collectively, these companies form the backbone of the \$2 trillion global electronics industry, which supports greater than 350,000 high-skill, high-wage jobs in the United States.

Semiconductors are fundamental to modern technology, found in everything from smartphones and computers to medical devices and defense systems. Many of these devices and services rely on chips that are specifically designed and manufactured to enable increasingly sophisticated AI capabilities. The United States currently possesses market and technological leadership in the design, manufacturing equipment, and materials that enable AI chip production, but the rise of global competitors and the offshoring of critical manufacturing processes have led to an erosion of U.S. preeminence in semiconductors and posed challenges to the stability of the U.S. semiconductor supply chain. As a consequence of this shift, the U.S. share of global semiconductor manufacturing has fallen from 37% in 1990 to just 12% in 2022.

While the United States remains the global leader in the design and development of AI chips in particular, recent events have highlighted the fact that this primacy is not guaranteed. This includes the unexpected success that overseas competitors have had in developing functionally equivalent but significantly less expensive AI systems despite restrictive export controls on advanced U.S. chip technology. As such, SEMI members urge the Administration to craft an AI Action Plan that makes a strong commitment to domestic semiconductor production through funding and tax incentives, innovation through publicly and privately funded research and development (R&D), a regulatory regime

that preserves access to critical chemicals and materials, the implementation of trade policies that allow for robust and fair competition in a global marketplace, and a domestic workforce development enterprise that helps ensure the availability of top-level talent.

To ensure that the United States maintains its competitive edge in AI, SEMI members make the following recommendations, which will be expanded upon in this document:

- **CHIPS Implementation** – Continue investing in the reshoring of semiconductor manufacturing through the CHIPS Program and the fielding of R&D initiatives relevant to AI.
- **Trade and Tariff Policy** – Maintain market access for semiconductor equipment manufacturers by leveraging strategic trade policies that promote U.S. technological leadership, enable fair competition, and avoid blanket measures that could undermine key industries.
- **Export Controls** – Support balanced export controls that protect national security while maintaining global competitiveness as nations vie for AI supremacy.
- **Tax Policy** Expand tax incentives to strengthen domestic semiconductor manufacturing.
- **Workforce Development** – Address workforce shortages through STEM education funding and targeted adjustments to the high-skilled immigration system.
- **Environmental and Energy Considerations** – Advocate for the enablement of environmentally efficient technologies that strengthens global competitiveness by driving innovation and supporting initiatives that aim to reduce excessive regulatory burdens.

CHIPS Act Implementation

As originally envisioned in the first Trump Administration to onshore critical semiconductor manufacturing capacity to protect national security and promote supply chain resiliency in the United States, the *Creating Helpful Incentives to Produce Semiconductors (CHIPS) for America Act* and the *CHIPS and Science Act* have collectively sought to level the international playing field by incentivizing domestic production of critical chips and enabling U.S. leadership in R&D and advanced packaging. This bipartisan effort has yielded the largest U.S. semiconductor expansion in decades and, when fully, effectively, and decisively administered, will continue to build upon U.S. leadership in semiconductor technology. These programs are particularly important for the nation's efforts to maintain global AI leadership. Specifically, bolstering domestic manufacturing of advanced AI chips will provide the United States with more secure access to a commodity that is growing in necessity as AI-enabled devices and systems increasingly pervade and enable modern life.

R&D is also foundational to the semiconductor industry's ability to design, develop, and manufacture the most advanced semiconductors as well as to field the high-skilled workforce that can sustain it in the long-term. The *CHIPS for America Act* recognized this reality when it authorized the establishment of several targeted R&D activities aimed at ensuring that the United States remains the global epicenter in developing new semiconductor technologies. These include the following programs, each of which are directly relevant to advancing AI leadership:

- **National Semiconductor Technology Center (NSTC)** – The NSTC is a public-private partnership that aims to accelerate innovation in semiconductors and their manufacturing process. Specifically, NSTC-led activities will focus on addressing challenges in areas that currently impede the development of more capable AI systems, including processing power and energy efficiency.

The NSTC is a unique entity in the United States, as there is no equivalent organization serving as a central hub for fostering multi-sector collaboration across the entire semiconductor ecosystem.

- **National Advanced Packaging and Manufacturing Program (NAPMP)** – The NAPMP is intended to expand domestic capacity in advanced packaging, a segment of the semiconductor manufacturing process for which the U.S. share is only three percent globally. Beyond the economic benefit of capturing a greater percentage of this market, bolstering domestic semiconductor packaging is critical to supply chain resilience. This is especially important given the increasing salience and pervasiveness of AI chips.
- **CHIPS Manufacturing USA Institute** – The CHIPS Manufacturing USA Institute, which will join a network of existing institutes, will focus on addressing challenges in semiconductor manufacturing using digital twins. These digital twins are computer models that can replace the need for physical objects, thereby reducing the cost of experimentation in advanced manufacturing. Digital twins can be augmented with AI to further optimize the semiconductor manufacturing process.
- **Metrology R&D** – The CHIPS Metrology R&D program will support measurement science for the development of new materials, packaging, and production methods for semiconductors. The increasingly small feature size of advanced AI chips necessitates ultra-precise measurement and inspection capabilities to ensure their reliability.

To maximize the impact of the *CHIPS for America Act* and *CHIPS and Science Act*, administering the direct company incentives program and implementing the R&D initiatives is essential, while eliminating burdensome non-statutory provisions. A stable, predictable, and efficient distribution system without significant delay will empower semiconductor companies to continue to invest confidently in new U.S. manufacturing and R&D facilities, driving long-term innovation and competitiveness, including many smaller-scale suppliers.

Key Policy Concerns:

1. **Overly Burdensome Regulatory Requirements** – Complex and burdensome application processes, extensive environmental reviews, and regulatory hurdles have reportedly discouraged some small and medium-sized companies with new and innovative technologies from applying and may have slowed the timeline for project approvals.
2. **Potential Rescoping of Implementation** – The current review of the CHIPS Program Office (CPO) by the new Administration allows for clarification of funding priorities, particularly in ongoing and future R&D initiatives, ensuring sustained federal support for planned semiconductor investments. Additionally, identifying creative solutions to encourage partnerships to strengthen workforce development programs will further enhance job growth and talent retention in the semiconductor industry.
3. **Encouraging Long-Term Semiconductor Roadmapping** – Developing an industry-led strategy that leverages federal tools to foster a stable, long-term investment climate will be an enabler in encouraging semiconductor companies to expand their manufacturing and R&D footprint in the United States. A coordinated and comprehensive approach—one that prioritizes pro-business

and pro-innovation policies—will secure U.S. technological leadership, strengthen supply chain resilience, and maximize returns for the American taxpayer.

Recommended Policy Actions:

- Streamline any remaining CHIPS funding application processes to accelerate approvals and lower the barrier of entry for small and medium enterprises.
- Ensure the CHIPS Program maintains its full intended scope, as authorized in statute, without additional burdensome regulation or funding reductions.
- Create a long-term semiconductor investment roadmap to encourage further investment beyond the CHIPS Program.

Trade & Tariff Policy

The complexities of the semiconductor industry supply chain require U.S. companies to rely on highly specialized materials, equipment, and components sourced from multiple countries. Strategically deployed and comprehensively assessed trade actions ensure that U.S. businesses maintain market access and remain globally competitive. Through fair and reciprocal trade practices, the U.S. can avoid unintended cost increases, supply chain disruptions, and ultimately any risks to U.S. competitiveness in the global marketplace.

Coordinating trade policies with our allies prevents retaliatory tariffs for industries critical for national security, such as the semiconductor industry, and keep costs competitive for consumers and industries, all while fostering a more resilient domestic manufacturing ecosystem. As demand for emerging technologies grows, empowering U.S. semiconductor companies to compete on a level playing field will help maintain U.S. innovation prowess for everything from consumer electronics to automotive and defense applications. A resilient, secure, and open global supply chain for the semiconductor industry both benefits U.S. economic interests and national security.

Recommended Policy Actions:

- Encourage the adoption of clear and targeted trade policies that encourage continued market access for U.S. companies.
- Support the negotiation of exemptions for industries critical for maintaining national security objectives with partner countries.
- Promote coordinated and targeted efforts to encourage the reshoring of domestic manufacturing.

Export Controls

Export controls are an essential tool to protect sensitive U.S. semiconductor technologies, including advanced AI chips, from adversarial nations while ensuring America maintains its technological edge. To be effective, controls should be targeted and narrowly tailored to an identified national security risk, closing loopholes that enable the safe transfer of technology while avoiding overly broad and unilateral measures that could cede market share to non-U.S. competitors by imposing disproportionate

restrictions to American companies. A coordinated approach—guided by industry insights and aligned with key allies—will strengthen enforcement, promote compliance, and protect U.S. innovation without undermining global competitiveness.

Key Policy Concerns:

1. **Loss of Market Access for U.S. Companies** – Trade policies informed by comprehensive analyses and industry participation will ensure that U.S. firms remain competitive in a global marketplace.
2. **Regulatory Complexity and Compliance Burdens** – Recent export control actions, including expansion of the Foreign Direct Product Rule (FDPR), have increased regulatory costs and uncertainty for semiconductor companies. Complexity of recent rules and rapid regulatory changes—without sufficient stakeholder input—create confusion, undermine companies’ ability to comply with export control rules, and limit the ability of businesses to plan long-term.
3. **Coordination with Like-minded Allies** – Plurilateral agreements on export controls with key allies will create a more cohesive regulatory framework, strengthening both national security and U.S. competitiveness. Coordinated policies help U.S. firms compete on a level playing field with competing firms from allied nations while supporting a resilient and interconnected global supply chain.

Recommended Policy Actions

- Refine export control rules to focus only on truly sensitive semiconductor technologies.
- Coordinate export control policies with allied nations and avoid unilateral export controls to ensure the competitiveness of U.S. companies abroad.
- Increase transparency in export control decision-making to minimize uncertainty for semiconductor companies across the supply chain.

Tax Policy and Investment Incentives

With most key tax provisions from the *2017 Tax Cuts and Jobs Act* set to expire at the end of the year and the *CHIPS and Science Act* Investment Tax Credit only applying to investments made before January 1, 2027, 2025 is expected to bring a high-stakes tax debate on Capitol Hill, shaping the future of U.S. semiconductor investment and broader economic policy. The high cost of semiconductor manufacturing in the U.S. is a significant challenge and a key driver behind the offshoring of critical aspects of semiconductor manufacturing.

Semiconductor infrastructure—especially the manufacturing facilities needed for producing the most advanced AI chips—requires billions of dollars in upfront investment, and tax incentives are essential to offset these costs. A competitive tax environment encourages semiconductor companies to invest in the United States, strengthening domestic manufacturing and innovation while maintaining global leadership against lower-cost regions. Increasing federal R&D tax incentives will also allow semiconductor firms to invest more in AI chips, quantum computing, and advanced chip architectures.

Key Benefits of Tax Incentives:

1. Encouraging Domestic Semiconductor Manufacturing
 - a. U.S. construction costs are significantly higher than in Asia, due to higher labor, acquisition, and operational expenses.
 - b. Investment tax credits (ITCs) help reduce costs, making the United States a more attractive location for semiconductor expansion.
2. Strengthening R&D and Innovation
 - a. Semiconductor companies invest over 15% to 20% of their annual revenue in R&D.
 - b. R&D tax credits drive technological advancements by allowing companies to reinvest savings into next-generation chip development.
3. Expanding the Semiconductor Supply Chain
 - a. Current tax incentives primarily support semiconductor fabs, but upstream suppliers also need incentives to strengthen the entire supply chain.
 - b. Expanding tax credits to include wafer production, chemical suppliers, and packaging will enhance the U.S. semiconductor ecosystem.

Recommended Policy Actions:

- Extend tax credits beyond fab construction to include semiconductor material suppliers and packaging facilities.
- Make *CHIPS and Science Act* tax credits permanent to provide long-term investment stability.
- Ensure tax incentives remain stable to encourage continued semiconductor investment.
- Strengthen R&D tax policies to promote next-generation semiconductor technology development.

Workforce Development & Immigration

One of the biggest headwinds facing U.S. semiconductor expansion is the shortage of skilled workers. Without a robust workforce, the semiconductor ecosystem currently being developed in the United States will struggle to find enough talent. This is especially true for the AI sector, where intensifying international competition has spurred a global scramble for talent.

Key Workforce Challenges:

1. Shortage of Skilled Workers – By 2030, the U.S. semiconductor industry will face a shortfall of over 300,000 engineers and technicians. Many semiconductor jobs require specialized expertise, but the United States lacks a national semiconductor workforce training strategy.
2. Immigration Barriers Restrict Talent Access – The H-1B visa program is capped at 85,000 visas annually, limiting the ability of semiconductor firms to hire top global talent as they compete against other industry for these specialized visas. Many U.S. semiconductor graduate students are foreign-born, yet immigration policies make it difficult for them to stay in the United States after graduation, benefitting the pools of talent in competing economies.

3. Lack of STEM Education & Apprenticeship Programs – Federal investment in semiconductor-specific STEM education remains low. Community colleges and technical schools play a key role in training semiconductor technicians but need greater funding and industry partnerships.

Recommended Policy Actions:

- Establish a national semiconductor workforce pipeline to align higher education, training, and industry needs, including through support of the National Network for Microelectronics Education authorized by the *CHIPS and Science Act*.
- Extend the H-1B grace period for finding a new employment sponsor from 60 days to 180 days to account for historical cyclicalities in advanced manufacturing.
- Expand apprenticeship programs in semiconductor manufacturing and engineering.
- Ensure that CHIPS Act funding continues to include workforce development support.

AI Resource Utilization

Fostering both a supportive regulatory environment and targeted R&D are essential to advancing resource-efficient technologies that enhance the global competitiveness of the U.S. semiconductor industry. This is especially important in AI, as the Department of Energy (DOE) has estimated that energy consumption from data centers—the infrastructure underpinning AI systems—tripled over the last decade and may triple again by 2028.¹ Additionally, the power consumption of semiconductor devices has been increasing over the last decade, leading to a dynamic wherein energy infrastructure is not keeping up with the demands of our growing reliance on AI. Finally, semiconductor manufacturing itself is also resource-intensive, requiring significant amounts of both energy and water.

As such, industry-led innovations in energy-efficient fabs, water recycling systems, and emissions reduction technologies are key to sustainable growth across the world. Targeted policies and investments in next-generation solutions will enable U.S. companies to account for energy needs while maintaining global competitiveness.

Recommended Policy Actions:

- Develop industry-led environmental standards for water recycling and emissions reduction.
- Expanding energy diversification to support and enhance the already energy intensive semiconductor manufacturing process.
- Leverage DOE R&D activities authorized under the Micro Act² to produce transformational outcomes in energy efficiency that will contribute to the more downstream research efforts of the NSTC and other CHIPS R&D programs.

¹ <https://eta-publications.lbl.gov/sites/default/files/2024-12/lbnl-2024-united-states-data-center-energy-usage-report.pdf>

² <https://www.energy.gov/science/articles/departments-energy-announces-179-million-microelectronics-science-research-centers>

- Build on the above R&D efforts by launching an NSTC technical center³ focused on sustainable semiconductor manufacturing.⁴

PFAS Regulations and Associated R&D

Creating semiconductors is a complex process, involving thousands of intricate steps. State-of-the-art technology nodes require over 2,000 processing steps, utilizing specialized equipment and materials, all of which collectively cost billions of dollars. Due to their extremely small feature size, chips are highly sensitive to any source of contamination and require atomic precision in manufacturing. These devices are only becoming more complex, and their feature sizes smaller, as demand for more powerful AI capabilities increases.

Various types of PFAS are essential to the semiconductor manufacturing process because they have low surface tension, high heat and chemical resistance, high thermal stability, radiation stability, favorable electrical characteristics, compatibility with other chemicals, purity, and other properties that enable them to fulfill the unique demands of semiconductor manufacturing. Fluoropolymers, fluorochemicals, and fluorinated gases are all used by the semiconductor industry to meet many needs within the manufacturing process.

Concerns about the pervasiveness of PFAS and their potential public health impacts have prompted policymakers at the federal and state levels to design broad regulations that often have unanticipated consequences. While the industry supports responsible PFAS management, overly broad regulations could limit access to essential materials without viable alternatives, potentially hindering innovation. SEMI advocates for balanced policies that allow continued PFAS use in critical applications while promoting research into next-generation materials and advanced manufacturing processes. Encouraging innovation through risk-based regulation and targeted R&D will strengthen the industry's ability to develop cutting-edge technologies that sustain semiconductor industry growth on the global stage.

Recommended Policy Actions:

- Support PFAS policies that empower authorities to implement risk-based analysis and emission controls while ensuring access to chemicals that are vitally important to the production of semiconductors.
- Leverage CHIPS R&D activities, such as the CARISSMA⁵, which aims to utilize AI to identify viable alternatives to PFAS and other chemicals currently used in semiconductor manufacturing, and PRISM⁶, which is intended to accelerate progress on environmental controls.

³ According to the [NSTC strategic vision](#) issued by the CPO, NSTC technical centers are designed to provide “end-to-end fabrication to enable small prototyping and pilot runs, experimentation with and testing of new materials and equipment, and other research-related activities.”

⁴ The [NSTC Strategic Plan](#) developed by CPO and Natcast, the NSTC operator, notes that “[sustainability] in semiconductor chip manufacturing is a broad topic including energy efficiency of the fab, water use, and emissions (greenhouse gases and toxic substances).”

⁵ <https://www.nist.gov/chips/r%2526d-funding-opportunities/notice-funding-opportunity-carissma>

⁶ <https://natcast.org/research-and-development/prism>

Conclusion

The United States has the resources, talent, and technological prowess to maintain global semiconductor supremacy, and therefore, leadership in AI. In service of this goal, SEMI recommends that the Administration's AI Action Plan communicate a bold, long-term, and unwavering commitment to the policy goals outlined in this RFI response. Specifically:

- Policymakers must ensure that implementation of the CHIPS Program continues and that it above all else prioritizes domestic manufacturing of advanced semiconductors, legacy semiconductors, and the associated supplier ecosystem.
- The Federal Government must work with industry to support R&D, workforce development, and advanced semiconductor manufacturing capabilities.
- The United States must work closely with allies to ensure a stable, secure, and globally competitive semiconductor supply chain.

SEMI stands ready to work with the Administration to ensure that the U.S. semiconductor industry and the AI leadership it enables remain strong, competitive, and future-focused.